Exploratory Data Analysis (EDA) Techniques

Simple-English reference tables: definitions, purpose, decisions, Titanic examples and visual forms

1. Univariate Analysis — one variable at a time

Use these techniques to understand a single column, including its typical value, spread, distribution and unusual values.

Technique	Definition	Purpose	What decisions can we make?	Example (Titanic dataset)	Visual
Mean	The average of all numerical values.	Understand the average value of a numerical column.	Understand the typical average and compare it with other groups or datasets.	Average passenger age is about 29 years.	A single number showing the average.
Median	The middle value after values are sorted.	Find the centre of the data without being strongly affected by extreme values.	Use the median instead of the mean when data is skewed or contains extreme values.	The median ticket fare is about $14.	A marked middle point on a sorted number line.
Mode	The value or category that appears most often.	Find the most common value.	Identify the most common category or value.	The most common passenger class is 3rd class.	A frequency bar for the most common category.
Minimum	The smallest value in a column.	Find the lowest observed value.	Check the lower end of the data and investigate impossible or unusual values.	The youngest passenger was about 0.42 years old.	The left/lower endpoint of a range.
Maximum	The largest value in a column.	Find the highest observed value.	Check the upper end of the data and investigate unusually large values.	The oldest passenger was 80 years old.	The right/upper endpoint of a range.
Standard deviation	Measures how far values typically spread from the mean.	Understand how much values vary.	Decide whether values are tightly grouped or widely spread; compare variability.	A larger age standard deviation means passenger ages vary more.	A dot plot or bell-shaped curve showing spread around the mean.
Frequency count	Counts how many times each value or category occurs.	See how often each category or value appears.	Find common/rare categories and notice possible class imbalance.	Count the number of male and female passengers.	A bar chart of category counts.
Histogram	Groups numerical values into intervals (bins) and shows the count in each interval.	See the shape and distribution of numerical data.	Look for skewness, peaks, gaps and possible unusual ranges.	Show how many passengers fall into each age range.	Adjacent bars for age intervals.
Box plot	Shows the median, lower and upper quartiles, whiskers and potential outliers.	Summarize spread and spot possible outliers.	Investigate unusual values and compare distributions between groups.	Check for unusually high passenger fares.	A box-and-whisker diagram.
IQR	The interquartile range: Q3 minus Q1; it covers the middle 50% of values.	Measure the spread of the middle half of the data.	Flag potential outliers using the 1.5 × IQR rule.	If Q1 = 10 and Q3 = 30, IQR = 20.	The length of the box in a box plot.
Z-score	Tells how many standard deviations a value is above or below the mean.	Compare a value with the overall distribution and flag unusual values.	Investigate values often flagged when z < -3 or z > +3; thresholds depend on context.	A fare with z-score +3.5 is unusually high relative to the mean.	A normal curve or number line with standard-deviation marks.

2. Bivariate Analysis — two variables together

Use these techniques to compare two variables or understand how they are related.

Technique	Definition	Purpose	What decisions can we make?	Example (Titanic dataset)	Visual
Correlation	Measures the direction and strength of a relationship between two numerical variables.	Check whether two variables tend to move together.	Identify positive, negative or weak linear relationships; correlation alone does not prove cause.	Explore the relationship between age and fare.	Scatter plot with an upward or downward trend.
Cross-tabulation (crosstab)	A table that counts combinations of two categorical variables.	Compare how categories occur together.	See which category combinations are more or less common; compare counts or percentages.	Count survived/not-survived passengers for each gender.	A two-way frequency table, sometimes shown as a heatmap.
Scatter plot	Plots each observation as a dot using two numerical variables.	See patterns, relationships, clusters and unusual points.	Decide whether a relationship or outliers need further investigation.	Plot passenger age against ticket fare.	Dots on x- and y-axes.
Line graph	Connects values in an ordered sequence with lines.	Show how a value changes across time or another meaningful order.	Spot increasing, decreasing or stable trends.	Plot passenger counts or survival rates across ordered age groups.	Points connected by a line.
Group comparison	Compares a measure or distribution across two or more groups.	Understand similarities and differences between groups.	Identify groups with higher/lower values and decide what to investigate next.	Compare survival rates for male and female passengers.	Side-by-side bars or box plots.
Average between categories	Calculates a numerical average separately for each category.	Compare typical numerical values across groups.	Find which category has the highest or lowest average; check group sizes and spread too.	Compare average fare for 1st, 2nd and 3rd class.	Bar chart of average fare by passenger class.

3. Multivariate Analysis — three or more variables

Use these techniques to explore relationships, patterns and groups across several features.

Technique	Definition	Purpose	What decisions can we make?	Example (Titanic dataset)	Visual
Correlation matrix	A table of correlation values for every pair of selected numerical features.	See many pairwise relationships in one place.	Find strongly related features, possible redundancy and potential multicollinearity; do not select features by correlation alone.	Compare Age, Fare, SibSp and Parch.	A colored heatmap with values from -1 to +1.
Pair plot	A grid of plots showing pairwise relationships between several numerical features; diagonal cells show each feature's distribution.	Explore many two-feature relationships together.	Spot patterns, possible clusters, outliers and distribution shapes.	View Age, Fare, SibSp and Parch together.	A grid of scatter plots and histograms/density plots.
Multiple-variable visualization	A chart that represents three or more variables at the same time, for example with position, color, size or shape.	Explore how several features interact.	Notice combinations of features associated with different outcomes.	Plot Age against Fare, color dots by survival and use marker shape for gender.	A scatter plot with color/shape encoding.
Principal Component Analysis (PCA)	Transforms correlated numerical features into fewer new variables called principal components.	Reduce dimensionality while retaining as much variation as possible.	Decide whether fewer dimensions make data easier to visualize or model; check retained variance and scaling.	Reduce several numeric passenger features to two components for a 2D view.	A 2D scatter plot with PC1 and PC2 axes.
Spatial analysis	Studies patterns using geographical location or spatial relationships.	Find where events or values occur and how they vary by place.	Identify geographic concentrations or regional differences when reliable location data exists.	Compare passenger outcomes by embarkation port; this is location-category analysis, not a detailed map unless coordinates are available.	A map or location-based chart.
Cluster analysis	Groups observations that are similar across selected features, without using known group labels to form the clusters.	Discover natural groups in the data.	Explore segments and decide whether groups are meaningful and useful; validate clusters before relying on them.	Group passengers using age, fare and family-size features.	A scatter plot with cluster groups shown in different colors.

4. Quick Guide — choosing an EDA technique

Start with the question you want to answer, then choose a technique that matches the data type and goal.

Question	Technique(s) to use
What is the average / middle / most common value?	Mean / Median / Mode
What are the smallest and largest values?	Minimum / Maximum
How much do values vary?	Standard deviation / IQR
How often does each category appear?	Frequency count
How is a numerical feature distributed?	Histogram / Box plot
Are there potential outliers?	Box plot / IQR / Z-score
How are two numerical features related?	Correlation / Scatter plot
How do two categorical features combine?	Cross-tabulation
How does a value change over an ordered sequence?	Line graph
How do groups differ?	Group comparison / Average by category
How are many numerical features related?	Correlation matrix / Pair plot
How do three or more features interact?	Multiple-variable visualization
Can many numerical dimensions be reduced?	PCA
Where do patterns occur geographically?	Spatial analysis
Which observations are similar?	Cluster analysis

Important notes

- IQR = Q3 – Q1. The common 1.5 × IQR rule flags values below Q1 – 1.5 × IQR or above Q3 + 1.5 × IQR as potential outliers.
- A common z-score rule flags values with an absolute z-score above 3 for investigation. This is a guideline, not an automatic instruction to delete values.
- Correlation shows association, not causation. A weak correlation does not always mean a feature is useless to a predictive model.
- PCA is sensitive to feature scales, so numerical features are commonly standardized first. Cluster results also depend on feature selection and scaling.
- Examples are illustrative; exact Titanic statistics can differ depending on the dataset version and how missing values are handled.